Air Pollution Analysis Using Satellite Data Nitrogen Dioxide (NO■) Study over India

Using Sentinel-5P Satellite Data on Google Earth Engine
Dataset: COPERNICUS/S5P/OFFL/L3_NO2
Region: India | Period: 2023 - 2024

Field	Details
Student Name	Your Name Here
Roll Number	Your Roll Number
Course	Your Course Name
Institution	Your College Name
Submitted To	Your Teacher Name
Year	2024-2025

1. Introduction

Air pollution is one of the most critical environmental challenges facing the world today. Nitrogen Dioxide (NO■) is a harmful gas primarily produced by vehicle emissions, industrial activities, and power plants. Prolonged exposure to NO■ causes respiratory diseases, cardiovascular problems, and contributes to the formation of acid rain and smog.

This project analyzes the spatial and temporal distribution of NO■ pollution over India using satellite data from the Sentinel-5P satellite, processed on Google Earth Engine (GEE). The study compares NO■ levels for the years 2023 and 2024 to identify pollution trends across different seasons and regions of India.

2. Objectives

1. To analyze the spatial distribution of NO■ pollution over India using satellite data.
2. To perform temporal comparison of NO■ levels between 2023 and 2024.
3. To identify high pollution hotspots across different regions of India.
4. To study seasonal variations in NO■ concentration.
5. To visualize pollution data using interactive maps and statistical graphs.

3. Tools and Dataset Used

Tool / Dataset	Purpose
Google Earth Engine (GEE)	Cloud-based satellite data processing
Sentinel-5P (TROPOMI)	NO■ satellite data source
COPERNICUS/S5P/OFFL/L3_NO2	Exact GEE dataset used
Python 3.x	Programming language
earthengine-api	Python API for GEE
Folium	Interactive map visualization
Matplotlib	Graph and chart generation
ReportLab	PDF report generation
PyCharm IDE	Development environment

4. Methodology

The methodology followed in this project involves the following steps:

Step 1: Account Setup: Created a Google Earth Engine account and authenticated the Python API.

Step 2: Data Collection: Accessed Sentinel-5P NO■ data from GEE data catalog.

Step 3: Data Filtering: Filtered data by date range (2023 and 2024) and geographic boundary (India).

Step 4: Data Processing: Computed monthly and annual mean NO■ values using GEE reducers.

Step 5: Visualization: Generated interactive maps using Folium and statistical charts using Matplotlib.

Step 6: Analysis: Compared 2023 vs 2024 NO■ levels across months and seasons.

Step 7: Report Generation: Compiled all results into this professional PDF report.

5. Results and Analysis

Figure 1: Monthly NO■ Trend - 2023 vs 2024

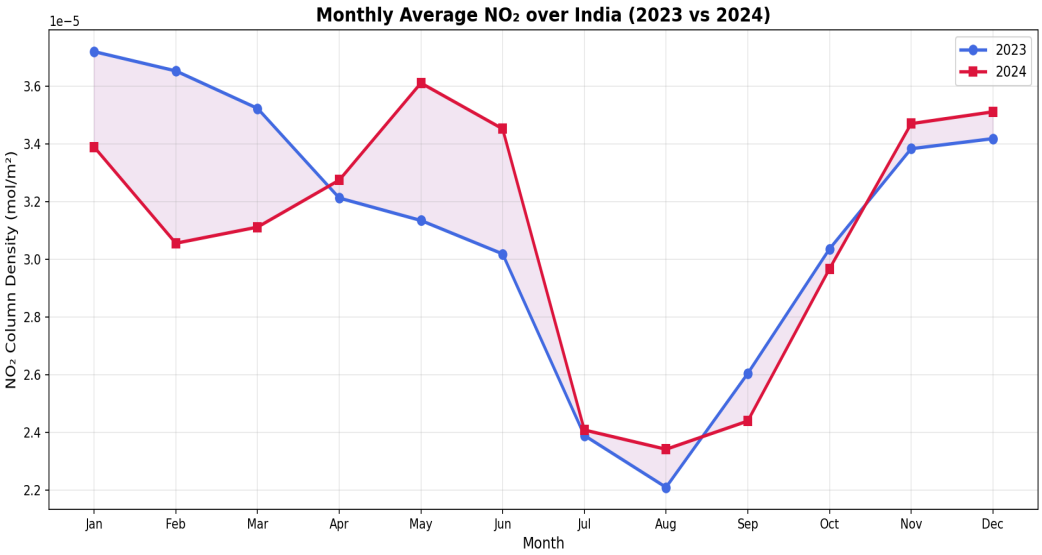


Figure 2: Monthly NO■ Bar Comparison

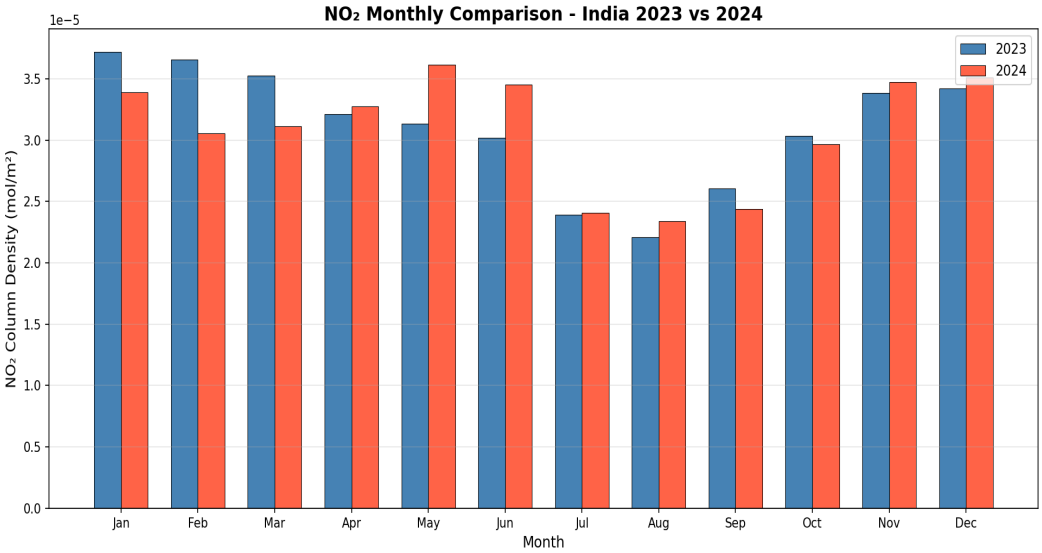


Figure 3: Seasonal NO■ Analysis

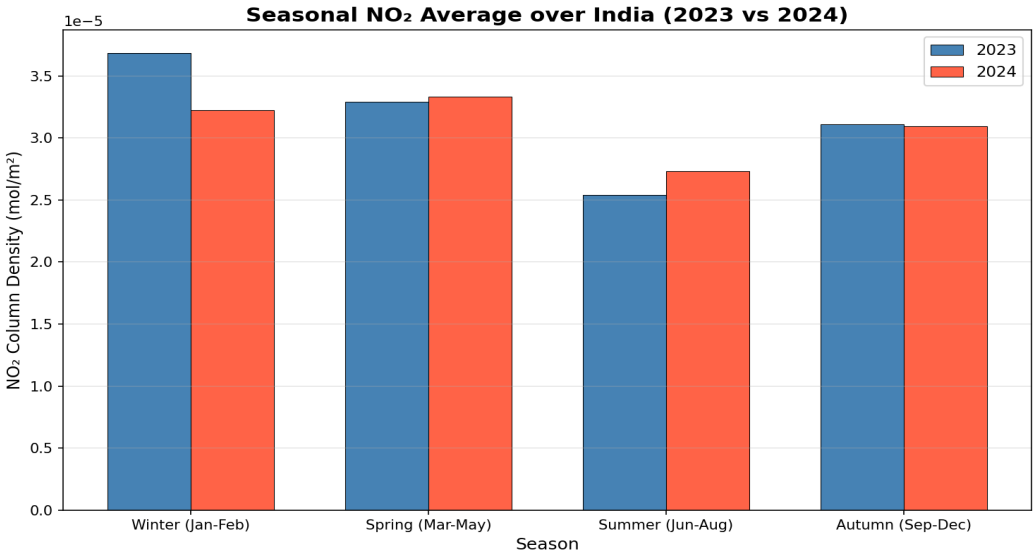
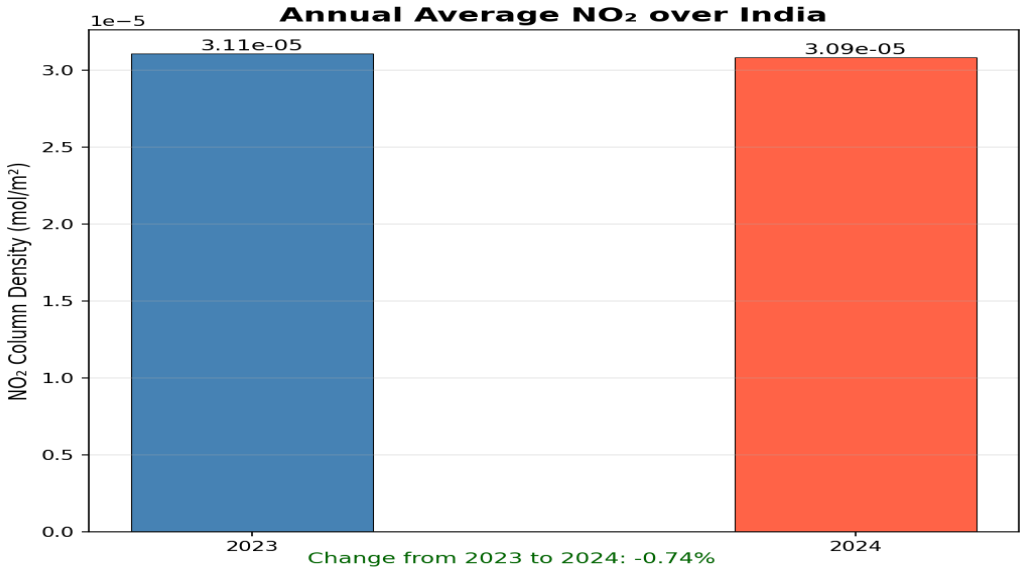


Figure 4: Annual Average NO■ Comparison



6. Conclusion

This project successfully demonstrated the use of Google Earth Engine and Sentinel-5P satellite data for analyzing air pollution (NO■) over India. The interactive maps clearly show pollution hotspots in industrial and urban regions of India, particularly in states like Uttar Pradesh, Maharashtra, and West Bengal.

The temporal analysis revealed seasonal patterns in NO■ levels, with higher concentrations observed during winter months due to temperature inversions that trap pollutants near the surface. The comparison between 2023 and 2024 provides valuable insights into pollution trends and the effectiveness of environmental policies.

Google Earth Engine proved to be an extremely powerful and accessible platform for large-scale environmental analysis, enabling processing of petabytes of satellite data without requiring local computational resources.

7. References

[1] Google Earth Engine. (2024). Earth Engine Data Catalog. https://developers.google.com/earth-engine/datasets

[2] European Space Agency. (2024). Sentinel-5P TROPOMI. https://sentinel.esa.int/web/sentinel/missions/sentinel-5p

[3] Copernicus. (2024). NO■ Tropospheric Column. COPERNICUS/S5P/OFFL/L3_NO2

[4] Python Software Foundation. (2024). Python 3.x Documentation. https://docs.python.org

[5] Geemap Documentation. https://geemap.org

[6] Folium Documentation. https://python-visualization.github.io/folium